Weather, Harvest & Risk — A Guide for the Farmer

This guide covers the three tabs that answer the question "*how is this season shaping up, and why?*" — **Weather**, **Analysis** (the harvest numbers), and **Risk** (the season risk score and harvest forecast). All three live on the read-only Owner View — the link ending `/owner/?key=...` that Admin issues under **Settings** → **Owner View**. They are not part of the Admin app itself, which keeps the day-to-day operational tabs.

> There is no tab literally called "Harvesting" — the harvest data lives under **Analysis**, and the harvest *prediction* lives under **Risk**. Both are covered here.

The last part of this guide — The risk model — explains in plain language where the score comes from, what it can and cannot tell you, and how to actually use it in a season.

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The Weather tab

What you're looking at

The farm's weather history, **1987 to today**, one line per year you tick. The data comes from a weather service (Open-Meteo) for the farm's own GPS position — set in Settings — not from an on-farm weather station. It is recorded hour by hour and then averaged or totalled into one point per day for the chart.

Every time you open the tab, the app quietly fetches any hours that have happened since the last time and adds them to the stored history. The line under the chart title — "*Weather data current to …*" — tells you how fresh it is. If the internet is down, the tab still shows everything already stored; it just won't be topped up.

The measurements

Tick as many as you like along the top:

Measurement	What the daily figure is
Temperature	The day's average
Humidity	The day's average
Dew Point	The day's average
Precipitation	The day's **total** rain, in mm
Wind Speed	The day's average
Soil Temp (6cm)	The day's average
UV Index	The day's **peak**, not its average
Sunshine	The day's **total** hours of sunshine

> **One gap worth knowing about:** Soil Temp and UV Index are only available from **2020 onward**. The long weather archive that reaches back to 1987 doesn't carry them, so those two lines simply stop for earlier years. Everything else runs the full record.

The years

Below the measurements, tick whichever years you want to compare. Every year is drawn on the same **1 January – 31 December** axis, laid over each other, so you can see directly whether this October was wetter than last October.

- The tab opens on the **most recent year on file**.
- Past years are drawn in **grey-blue** — the lighter the shade, the older the year.
- The **current year** is always drawn in a **red/orange** shade, and drawn slightly heavier, so the season in progress is never mistaken for history.

> Note the calendar here is a plain **calendar year**, not the harvest season. The Analysis tab, by contrast, runs its season from **1 August**. That's deliberate: "what was the weather like in 2023" naturally means the whole of 2023, while "the 2023 harvest" means the picking season.

Getting it off the screen

The small **PDF button** in the card's top-right corner downloads the chart exactly as it appears as a one-page PDF, headed with the farm name — ready to print, email, or drop into a report.

What it's good for

- Comparing this spring against previous springs, before the crop tells you.
- Checking the risk factors yourself. If the Risk tab says "Flowering-Period Sunshine" was poor this year, tick Sunshine and the last few years here and you can see it with your own eyes.
- Settling arguments about whether a season really was drier or hotter than people remember.

The Analysis tab — the harvest numbers

This is the harvest side of the same question: not what the weather did, but what actually came off the trees. It compares the **current season** against the daily historical records for **2020–2025**, which were imported once from the farm's own pre-app spreadsheets and never written to again.

At the top, four quick numbers: **Season to Date** (kg picked so far), **vs 5-Yr Average Pace** (green if ahead of where the 5-year average stood on this same day of the season, red if behind), the **Current Season** being compared, and how many **Years of History** are on file.

Below that, each card answers a different question:

- **Season Pace** — cumulative kg day by day from 1 August, this season in navy against every prior year plus a dashed 5-year-average line. Shows not just *whether* you're ahead or behind, but *when* the gap opened.
- **Per-Block Yield** — this season's kg per hectare (or per tree) for each block against that block's own history, with a farm-average line. Green or red in the **vs Average** column, so under-performing blocks stand out straight away.
- **Harvest Volume by Block and Season** — a bubble per block per season, sized by kg, blocks ranked by total volume.
- **Yield per Tree/Hectare by Block and Season** — a heatmap of every block against every season. Read across a row for one block's history, down a column for how a whole season went.
- **Variety Performance Over Time** — average kg per tree per variety, by season, for spotting a variety trending up or down.
- **Harvest Season Length** — one bar per season from first pick to last, labelled with the span and how many of those days actually had picking.
- **Monthly Harvest Volume** — kg by month for every season, with each month's share of that season's total, so a shift in *when* the crop comes in is visible.

Every one of these cards has the same PDF download button as the Weather chart.

> **A caveat on the historical block figures.** Blocks **8a/8b**, **10a/10b**, **17a/17b** and **19a/19b** didn't exist separately before the app — the old spreadsheets recorded one combined total for each pair. Those totals have been split between the two sub-blocks in proportion to their hectares. It's a fair estimate, not a record of what was actually picked from each half, and every figure built from a split carries a small info icon so it can't be mistaken for an exact number.

The risk model

What the score is

The **Critical Season Risk Indicator** is a score from **0 to 100** of how risky *this season's weather* looks for a poor harvest. Higher is worse.

It is built from **four weather factors that best explained this farm's own harvest variation** across its own record. It is **not** a generic lychee agronomy model, and it is **not** a weather forecast dressed up as a yield figure. Every number in it is anchored to seasons that actually happened on this farm.

The score falls into one of four bands:

Score	Band
below 25	**Low**
25 to below 50	**Moderate**
50 to below 75	**Elevated**
75 and above	**High**

Where the four factors came from

The four factors are the result of a **one-off study**, not something the app recalculates every time you open it. That matters: if the app re-picked its "best" factors each season, the meaning of the score would drift and you could never compare one year's score to another's.

The study was last re-run once the farm's full **1987–2025** weather and harvest record had been imported. It tested roughly **230 combinations** of weather measure and time window against three separate stretches of the farm's history:

- **Block 7 alone, 1987–2025** (37 seasons) — the one block continuously in production across the whole record; the old workbook's "Blok 7" is today's block 7, the same 2,132 trees planted in 1982.
- **The old orchard's whole-farm total, 1987–2009** (23 seasons).
- **The replanted orchard, 2012–2025** (14 seasons).

Only factors that **pointed the same way in all three** were kept, and they were also checked against **each other** so the score isn't counting one underlying signal four times over. That last check threw out a lot: dew point, humidity, dry hours and day-night temperature swing all measure much the same thing — spring dryness — and move together so closely that any one of them stands in for the rest. Only one of them survives in the score, and the four that were kept overlap with each other very little.

> An earlier version of this score, fitted on 2020–2025 alone, included **winter chill hours** and **rain days during flowering**. Once there was enough history to test them properly, both turned out to have no relationship with this farm's yields at all — winter chill correlated at essentially zero across 37 seasons — and they were replaced.

The four factors

Each is measured over a **fixed calendar window** — the same dates every year, regardless of when picking actually starts — and contributes up to **25 risk points**.

Factor	Window	What's measured	Which way is bad
Fruit Development Air Dryness	16 Sep – 31 Oct	Average dew point (°C)	**Lower** (drier air) is worse
Fruit Development Warmth	16 Sep – 15 Nov	Average of each day's *maximum* temperature (°C)	**Higher** is worse
Flowering-Period Sunshine	1 Aug – 15 Sep	Total hours of sunshine	**Lower** is worse
Fruit-Sizing Rainfall	1 Oct – 30 Nov	Total rain (mm)	**Lower** is worse

Why each one:

- **Air dryness during fruit development** is the single strongest weather signal in the whole record. Dry air while the fruit is sizing means the crop loses water faster than the tree can take it up, and those seasons came in smaller. Dew point measures that dryness more directly than humidity on its own.
- **Warmth during fruit development** — persistently warm afternoons through this period line up with smaller crops here. It's measured as the average of each day's peak rather than a flat average across day and night, so it picks up a genuinely hot spring rather than being washed out by cool nights. It's also a steadier measure than counting rare extreme days.
- **Sunshine during flowering** — bright flowering weather produced the bigger crops in this farm's record. Dull, overcast spells while the trees are in flower mean poorer pollination and fruit set.
- **Rain while the fruit fills out** fed this farm's bigger harvests; the dry Octobers and Novembers line up with the smaller ones.

How a season is scored

Each factor is scored on a sliding scale between the **best and the worst value seen across the reference seasons — 2012 onward**:

- The best value ever recorded for that factor scores **0 points.**
- The worst value ever recorded scores **25 points.**
- Anything in between scales proportionally.
- Anything *beyond* either extreme is held at 0 or 25 rather than pushed further.

So **25 out of 25 means "as bad as the worst season we have on file for this factor"** — it is not an absolute agronomic threshold. The four scores are added up to give the 0–100 total.

> **Why 2012?** That's the first season of the replanted orchard with per-block records. Going further back would mix in the *old* orchard, which was grubbed and replanted around 2010 — a completely different set of trees.

Windows that haven't closed yet are left out entirely. They are never assumed to be zero risk. This is why the current season shows a greyed **"Score so far"** with a count — "*2 of 4 factors known*" — instead of a final number, until the last window closes on 30 November.

The **Season** dropdown lets you look at any past season the same way, with its own score, band, and a card per factor showing that season's actual value against the reference range and how many of the 25 points it earned.

The Harvest Forecast card

At the top of the Risk tab, the same four factors are turned into **three kg predictions for the current season — Favorable**, **Expected** and **Unfavorable** — rather than one falsely-precise number, since most of a season's outcome still depends on weather that hasn't happened yet.

How each factor's window is filled in. For any window not yet closed, the app splits the remaining days three ways, and the **Basis** column in the table shows you exactly how the split fell (e.g. "*14d actual + 15d forecast + 16d assumed*"):

1. **Actual** — days already past, using real recorded weather.
2. **Forecast** — up to **15 days ahead**, using a real short-range weather forecast.
3. **Assumed** — everything beyond that, filled in from the historical range. This is where the three scenarios differ: **Favorable** assumes the best value seen in any reference season, **Expected** the average, **Unfavorable** the worst.

Each scenario's projected weather is then scored through the **exact same 0–100 scoring** used for real seasons, so a scenario's score is directly comparable to any past year's.

Turning a score into kilograms. The score is converted to kg by a straight line fitted through the (risk score, harvest total) pairs from **2016 onward.** That's a narrower range than the score's own

2012 reference, and deliberately so: in 2012–2015 block 7 was the only block bearing at all — the replanted blocks produced literally nothing until 2016 — so those small totals reflect a young orchard, not bad weather. Including them would teach the line that a middling risk score means a tiny crop. The line's strength and how many seasons it was fitted on are both shown in the methodology panel.

The two extremes are held to what has actually happened. Predictions are clamped to the best and worst harvests on record rather than letting the line run out past them (at the unfavorable end, it otherwise ran straight past zero into negative kg). So read the outer two cards as "*about as good, or as bad, as it has ever gone here*" — not as exact figures.

> **The Unfavorable scenario is harsher than any real season.** It combines each factor's own worst historical *year* — four different years, not one real season that went wrong on everything at once. Because the four factors are close to independent, a season that is simultaneously at its worst on all four is very unlikely. Treat it as a floor to plan against, not as a probable outcome.

If the live forecast can't be reached, the card still appears — every factor simply falls back to the historical range for its whole remaining window, and a note says so. Likewise, a genuine gap in the recorded data (a service outage) makes that factor fall back to the historical range for its entire window rather than quietly treating missing data as "no risk"; that row is flagged in the table.

The Risk tab makes a live call out to the weather service and reads the whole history, so it can take a few seconds to fill in. The "*Working out this season's forecast…*" message means it's still going, not that something is broken.

Checking it against reality

Below the season inspector, two bar charts sit stacked deliberately: **Risk Score by Season** and **Actual Harvest by Season**, same years, same order. This is the back-test — you can check by eye whether the seasons that scored badly really did come in smaller. It is there so the score can be judged, not just believed. The current season is marked with an asterisk and may still be partial.

What the score cannot tell you

These limits are printed in the app itself, under "*How this score is calculated*", and they matter:

- **Weather explains roughly a quarter to a half of this farm's swing from season to season.** That is real signal — but most of what makes a season good or bad is something this score simply cannot see: pruning, nutrition, irrigation, pests, labour, timing of the pick.
- **Around 230 combinations were tested to arrive at these four.** In any search that wide, some of what looks like signal is chance. These four were kept because they held up across separate stretches of the farm's history, not because they scored well once. The three stretches also overlap — block 7 is a large share of the modern farm's total — so they aren't three fully independent confirmations.

- **There is no alternate-bearing pattern on this farm.** Growers often expect lychees to alternate a big year with a small one. Across 35 back-to-back season pairs here, last season's crop tells you essentially nothing about this one. (An earlier version of the app claimed the opposite; the longer record disproved it.)
- **It says nothing about quality, size, or price** — only about total kg.

How to use it through a season

The score fills in on a fixed timetable. Because the windows are fixed calendar dates, you always know when each factor locks in:

From	What you know
1 Aug	Nothing final yet — flowering sunshine window opens
16 Sep	**1 of 4 —** Flowering-Period Sunshine is final
1 Nov	**2 of 4 —** Fruit Development Air Dryness is final
16 Nov	**3 of 4 —** Fruit Development Warmth is final
1 Dec	**4 of 4 —** the season's score is final

So the score is at its least certain exactly when it would be most useful, and fully certain only once the crop is largely determined. That is honest, and it's the reason for the three-scenario forecast: early in the season, **the spread between Favorable and Unfavorable is the real information**, not the Expected figure in the middle.

Practical use:

- **Plan against the range, not the middle.** In August and September, look at the gap between Favorable and Unfavorable and make sure both ends are survivable — crates, labour, transport, packhouse slots, cash flow. As the season closes the windows one by one, that gap narrows and you can commit harder.
- **Look at *which* factor is scoring badly, not just the total.** The per-factor cards tell you where the risk is coming from, and two of the four are things you can partly answer:
- **Air dryness** and **fruit-sizing rainfall** are both water. A season scoring high on these is a season to bring irrigation forward and be less sparing with it.
- **Flowering sunshine** and **fruit-development warmth** you cannot do anything about — but knowing early that they've gone against you is worth having when you're deciding how much to commit elsewhere.
- **Use it as one input among several.** Walk the orchard, look at fruit set, and read the **Season Pace** chart in the Analysis tab once picking starts. If the pace chart and the risk score disagree, believe the fruit on the trees — the pace chart is measuring what actually happened, and the score is only measuring the weather's contribution to it.

- **Look back before you trust it forward.** Use the Season dropdown to pull up two or three past seasons you remember well, and see whether the score matched what you lived through. That's the fastest way to calibrate how much weight to give it.

What not to do with it:

- Don't quote the Expected kg figure as a commitment. It's a description of what the historical pattern implies about this season's weather — not a promise of what will be harvested.
- Don't compare this score to another farm's. Every number in it is scaled against *this* farm's own best and worst seasons.
- Don't read last season's score as telling you anything about this one. The record here says it doesn't.